

DEPARTMENT OF MATHEMATICS, IIT Guwahati
MA102: Mathematics II
Tutorial Problems Sheet

11. Examine whether the following sets are subspaces of $\mathbb{R}^n$ ($n \geq 3$).

$$S_1 = \{[x_1, \dots, x_n]^t \in \mathbb{R}^n : x_1 + x_2 = 4x_3\}, \quad S_2 = \{[x_1, \dots, x_n]^t \in \mathbb{R}^n : x_1 + x_2 \leq 4x_3\},$$

$$S_3 = \{[x_1, \dots, x_n]^t \in \mathbb{R}^n : x_1 = 1 + x_2\} \quad \text{and} \quad S_4 = \{[x_1, \dots, x_n]^t \in \mathbb{R}^n : x_1 x_2 = 0\}.$$

Remark: If a set S is a subspace then show that the rules

Rule 1 : $\mathbf{0} \in S$ (to claim that $S \neq \emptyset$);

Rule 2 : $\alpha \mathbf{u} + \beta \mathbf{v} \in S$ for every $\mathbf{u}, \mathbf{v} \in S$ and $\alpha, \beta \in \mathbb{R}$.

are satisfied. If S is **not** a subspace then show that either $\mathbf{0} \notin S$ or show, *by a counter-example*, that **Rule 2** is not satisfied.

Solution: Clearly $\mathbf{0} \in S_1$ and hence $S_1 \neq \emptyset$. (**This is an important step.**) Let $\mathbf{u} = [x_1, \dots, x_n]^t, \mathbf{v} = [y_1, \dots, y_n]^t \in S_1$ and $a, b \in \mathbb{R}$. Then we have $x_1 + x_2 = 4x_3$ and $y_1 + y_2 = 4y_3$. Now

$$a\mathbf{u} + b\mathbf{v} = [ax_1 + by_1, ax_2 + by_2, ax_3 + by_3, \dots, ax_n + by_n]^t.$$

We have $(ax_1 + by_1) + (ax_2 + by_2) = a(x_1 + x_2) + b(y_1 + y_2) = 4(ax_3 + by_3)$. Hence $a\mathbf{u} + b\mathbf{v} \in S_1$ for any $\mathbf{u}, \mathbf{v} \in S_1$ and $a, b \in \mathbb{R}$, and therefore S_1 is a subspace of $\mathbb{R}^n$.

We have $\mathbf{u} = [1, 1, 1, \dots, 1]^t \in S_2$ but $(-1)\mathbf{u} \notin S_2$ since $-1 - 1 > -4$. Hence S_2 is not a subspace of $\mathbb{R}^n$.

We see that $\mathbf{0} \notin S_3$. Hence S_3 is not a subspace of $\mathbb{R}^n$.

We see that $\mathbf{e}_1 = [1, 0, 0, \dots, 0]^t, \mathbf{e}_2 = [0, 1, 0, \dots, 0]^t \in S_4$. However, $1\mathbf{e}_1 + 1\mathbf{e}_2 = \mathbf{e}_1 + \mathbf{e}_2 \notin S_4$. Hence S_4 is not a subspace of $\mathbb{R}^n$. $\square$

12. Find all the subspaces of $\mathbb{R}^2$.

Solution: Clearly, $\{\mathbf{0}\}$ is a subspace of $\mathbb{R}^2$. Let $S (\neq \{\mathbf{0}\})$ be another subspace of $\mathbb{R}^2$. Then there exists $\mathbf{u} = [u_1, u_2]^t \in S$ such that $\mathbf{u} \neq \mathbf{0}$. Notice that $\{\alpha \mathbf{u} : \alpha \in \mathbb{R}\} = \text{span}(\mathbf{u}) \subseteq S$. Now either $\text{span}(\mathbf{u}) = S$ or $\text{span}(\mathbf{u}) \subsetneq S$. In the first case, S is the straight line in $\mathbb{R}^2$ passing through the origin and $\mathbf{u}$.

If $\text{span}(\mathbf{u}) \subsetneq S$, then there exists $\mathbf{v} = [v_1, v_2]^t \in S \setminus \text{span}(\mathbf{u})$, and therefore $\mathbf{v} \neq s\mathbf{u}$ for any $s \in \mathbb{R}$.

Thus the set $\{\mathbf{u}, \mathbf{v}\}$ is linearly independent, *i.e.*, the rank of the matrix $A = \begin{bmatrix} u_1 & v_1 \\ u_2 & v_2 \end{bmatrix}$ is 2.

Therefore for any $\mathbf{b} = [b_1, b_2]^t \in \mathbb{R}^2$, we have $\text{rank}(A) = \text{rank}([A \mid \mathbf{b}])$, and hence the system $A\mathbf{x} = \mathbf{b}$ will have a solution $\mathbf{w} = [\alpha, \beta]^t$. Thus $A\mathbf{w} = \mathbf{b}$, which implies that $\mathbf{b} = \alpha \mathbf{u} + \beta \mathbf{v}$. Thus $\mathbb{R}^2 = \text{span}(\mathbf{u}, \mathbf{v}) \subseteq S$, and hence $\mathbb{R}^2 = S$.

Thus the subspaces of $\mathbb{R}^2$ are $\{\mathbf{0}\}, \mathbb{R}^2$ and the straight lines passing through the origin. $\square$

13. Show that $W = \{[x, y, z, w]^t \in \mathbb{R}^4 : w - z = y - x\}$ is a subspace of $\mathbb{R}^4$, spanned by the vectors $[1, 0, 0, -1]^t$, $[0, 1, 0, 1]^t$ and $[0, 0, 1, 1]^t$. Also, determine $\dim(W)$.

Solution: We have

$$\begin{aligned} W &= \{[x, y, z, w]^t \in \mathbb{R}^4 : w - z = y - x\} \\ &= \{[x, y, z, y - x + z]^t : x, y, z \in \mathbb{R}\} \\ &= \{x[1, 0, 0, -1]^t + y[0, 1, 0, 1]^t + z[0, 0, 1, 1]^t : x, y, z \in \mathbb{R}\} \\ &= \text{span}([1, 0, 0, -1]^t, [0, 1, 0, 1]^t, [0, 0, 1, 1]^t). \end{aligned}$$

Thus W is a subspace of $\mathbb{R}^4$, spanned by $[1, 0, 0, -1]^t$, $[0, 1, 0, 1]^t$ and $[0, 0, 1, 1]^t$.

Now $a[1, 0, 0, -1]^t + b[0, 1, 0, 1]^t + c[0, 0, 1, 1]^t = [0, 0, 0, 0]^t \Rightarrow a = b = c = 0$.

Thus the set $B = \{[1, 0, 0, -1]^t, [0, 1, 0, 1]^t, [0, 0, 1, 1]^t\}$ is linearly independent. Hence B is a basis for W , and consequently $\dim(W) = 3$. $\square$

14. Find three vectors in $\mathbb{R}^3$ which are linearly dependent but any two of them are linearly independent.

Solution: The idea is to take three vectors corresponding to three distinct straight lines through origin where all of them lie on the same plane. For example, take $\mathbf{u} = [1, 0, 0]^t$, $\mathbf{v} = [0, 1, 0]^t$ and $\mathbf{w} = [1, 1, 0]^t$. $\square$

15. Let $S = \{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4\}$, where $\mathbf{x}_1 = [1, 0, 0, 0]^t$, $\mathbf{x}_2 = [1, 1, 0, 0]^t$, $\mathbf{x}_3 = [1, 2, 0, 0]^t$ and $\mathbf{x}_4 = [1, 1, 1, 0]^t$. Determine all $\mathbf{x}_i \in S$ such that $\text{span}(S) = \text{span}(S \setminus \{\mathbf{x}_i\})$.

Solution: It is easy to observe that $\mathbf{x}_1 = 2\mathbf{x}_2 - \mathbf{x}_3$. Thus $\mathbf{x}_2 = \frac{1}{2}\mathbf{x}_1 + \frac{1}{2}\mathbf{x}_3$ and $\mathbf{x}_3 = 2\mathbf{x}_2 - \mathbf{x}_1$. Hence $\text{span}(S) = \text{span}(S \setminus \{\mathbf{x}_i\})$ for $i = 1, 2, 3$.

However, $\mathbf{x}_4 = a\mathbf{x}_1 + b\mathbf{x}_2 + c\mathbf{x}_3 \Rightarrow a + b + c = 1, b + 2c = 1, 0 = 1$, which is absurd.

Hence $\text{span}(S) \neq \text{span}(S \setminus \{\mathbf{x}_4\})$. $\square$

16. Let $\mathbf{u}, \mathbf{v}, \mathbf{w}$ be linearly dependent vectors in $\mathbb{R}^n$ and let $\mathbf{v}, \mathbf{w}, \mathbf{x}$ be linearly independent vectors in $\mathbb{R}^n$, where $n \geq 3$. Show that (i) $\mathbf{u}$ is a linear combination of $\mathbf{v}$ and $\mathbf{w}$, and (ii) $\mathbf{x}$ is not a linear combination of $\mathbf{u}, \mathbf{v}$, and $\mathbf{w}$.

Solution: (i) Since $\mathbf{u}, \mathbf{v}, \mathbf{w}$ are linearly dependent, there exist $a, b, c \in \mathbb{R}$, not all zero, such that $a\mathbf{u} + b\mathbf{v} + c\mathbf{w} = \mathbf{0}$. Now if $a = 0$, then we have $b\mathbf{v} + c\mathbf{w} = \mathbf{0}$, where $(b, c) \neq (0, 0)$. Therefore $b\mathbf{v} + c\mathbf{w} + 0\mathbf{x} = \mathbf{0}$, where $(b, c) \neq (0, 0)$. This implies that $\mathbf{v}, \mathbf{w}, \mathbf{x}$ are linearly dependent, a contradiction. Thus $a \neq 0$, and therefore $\mathbf{u} = \left(-\frac{b}{a}\right)\mathbf{v} + \left(-\frac{c}{a}\right)\mathbf{w}$.

(ii) Let, if possible, $\mathbf{x} = \alpha\mathbf{u} + \beta\mathbf{v} + \gamma\mathbf{w}$ for some $\alpha, \beta, \gamma \in \mathbb{R}$. In **Part (i)**, we have got that $\mathbf{u}$ is a linear combination of $\mathbf{v}$ and $\mathbf{w}$. Therefore $\mathbf{x} = \alpha\mathbf{u} + \beta\mathbf{v} + \gamma\mathbf{w}$ gives that $\mathbf{x}$ is a linear combination of $\mathbf{v}$ and $\mathbf{w}$. This implies that the vectors $\mathbf{v}, \mathbf{w}, \mathbf{x}$ are linearly dependent, a contradiction. Hence $\mathbf{x}$ is not a linear combination of $\mathbf{u}, \mathbf{v}$, and $\mathbf{w}$. $\square$

17. Let $M = \{[x, y, z]^t \in \mathbb{R}^3 : x + y + 4z = 0\}$ and $N = \{[x, y, z]^t \in \mathbb{R}^3 : x + y + z = 0\}$. Show that M and N are subspaces of $\mathbb{R}^3$. Determine a basis for each of $M, N, M + N$ and $M \cap N$. Also, interpret your result geometrically.

Solution: It is left to the students to show that M and N are subspaces of $\mathbb{R}^3$.

For any $[x, y, z]^t \in M$, we have $x = -y - 4z$. Hence $[x, y, z]^t = [-y - 4z, y, z]^t = y[-1, 1, 0]^t + z[-4, 0, 1]^t$. Also, $\{[-1, 1, 0]^t, [-4, 0, 1]^t\}$ is linearly independent (**Check!**).

Thus $\{[-1, 1, 0]^t, [-4, 0, 1]^t\}$ is a basis for M . Similarly, $\{[-1, 1, 0]^t, [-1, 0, 1]^t\}$ is a basis for N .

Since $[-1, 0, 1]^t \notin M$, we find that $\{[-1, 0, 1]^t, [-1, 1, 0]^t, [-4, 0, 1]^t\}$ is a linearly independent subset of $M + N$. Moreover, $\text{span}([-1, 0, 1]^t, [-1, 1, 0]^t, [-4, 0, 1]^t) = M + N$. Hence $\{[-1, 0, 1]^t, [-1, 1, 0]^t, [-4, 0, 1]^t\}$ is a basis for $M + N$.

For any $[x, y, z]^t \in M \cap N$, we have $x + y + 4z = 0, x + y + z = 0$, that is, $x + y = 0, z = 0$. Hence $\{[-1, 1, 0]^t\}$ is a basis for $M \cap N$.

Geometric Interpretation: The subspaces M and N represent two planes in $\mathbb{R}^3$ (which are not parallel), each passing through the origin. The subspace $M \cap N$ is the straight line in $\mathbb{R}^3$ passing through the origin and $[-1, 1, 0]^t$.

The basis $\{[-1, 0, 1]^t, [-1, 1, 0]^t, [-4, 0, 1]^t\}$ is also a basis for $\mathbb{R}^3$. Hence $M + N = \mathbb{R}^3$. Thus any point in $\mathbb{R}^3$ can be expressed as a sum of two points, one lying on the plane M and the other lying on the plane N . $\square$

18. Let $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a linearly independent set of a vector space V . Write $\mathbf{w}_1 = \mathbf{v}_1$ and $\mathbf{w}_k = \sum_{i=1}^k c_{ik} \mathbf{v}_i$ for $2 \leq k \leq n$ such that $c_{ik} > 0$ for all $1 \leq i \leq n, 2 \leq k \leq n$. Show that the set $\{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_n\}$ is also linearly independent.

Solution: Let $a_1 \mathbf{w}_1 + a_2 \mathbf{w}_2 + \dots + a_n \mathbf{w}_n = \mathbf{0}$. This gives that

$$(a_1 + a_2 c_{12} + \dots + a_n c_{1n}) \mathbf{v}_1 + (a_2 c_{22} + \dots + a_n c_{2n}) \mathbf{v}_2 + \dots + [a_{n-1} c_{(n-1)(n-1)} + a_n c_{(n-1)n}] \mathbf{v}_{n-1} + a_n c_{nn} \mathbf{v}_n = \mathbf{0}.$$

Since $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a linearly independent set, we have

$$\begin{aligned} a_1 + a_2 c_{12} + \dots + a_n c_{1n} &= 0 \\ a_2 c_{22} + \dots + a_n c_{2n} &= 0 \\ &\vdots \\ a_{n-1} c_{(n-1)(n-1)} + a_n c_{(n-1)n} &= 0 \\ a_n c_{nn} &= 0. \end{aligned}$$

The above system of equations gives that $a_n = a_{n-1} = \dots = a_2 = a_1 = 0$. Hence the set $\{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_n\}$ is also linearly independent. $\square$

19. Consider the vector space $\mathbb{R}_4[x]$ of all polynomials with real coefficients of degree at most 4. Show that $S = \{p(x) \in \mathbb{R}_4[x] : p(1) = p(-1) = 0\}$ is a subspace of $\mathbb{R}_4[x]$. Determine a basis and the dimension of this subspace.

Solution: Clearly, $0 \in S$. We know that if a is a root of the polynomial equations $p(x) = 0$ and $q(x) = 0$ then for any $\alpha, \beta \in \mathbb{R}$, a is a root of the polynomial equation $\alpha p(x) + \beta q(x) = 0$ as well. This gives that if $p(x), q(x) \in S$, $\alpha, \beta \in \mathbb{R}$, then $\alpha p(x) + \beta q(x) \in S$. Hence S is a subspace of $\mathbb{R}_4[x]$.

Note that the condition $p(1) = p(-1) = 0$ implies that $(1 - x^2)$ is a factor of $p(x)$. Thus $p(x)$ has the form $(1 - x^2)(a + bx + cx^2)$ for some $a, b, c \in \mathbb{R}$. From this, it is easily obtained that $\{(1 - x^2).1, (1 - x^2).x, (1 - x^2).x^2\}$ is a basis for S . $\square$

20. Let $M_n(\mathbb{R})$ denote the vector space of all $n \times n$ real matrices. Show that the set $S_n(\mathbb{R}) = \{A \in M_n(\mathbb{R}) : A = A^t\}$ is a subspace of $M_n(\mathbb{R})$. Also, find a basis and the dimension of $S_n(\mathbb{R})$.

Solution: Clearly, $\mathbf{O} \in S_n(\mathbb{R})$. We know that if A, B are symmetric matrices of the same size, then $\alpha A + \beta B$ is also symmetric for all $\alpha, \beta \in \mathbb{R}$. This gives that $\alpha A + \beta B \in S_n(\mathbb{R})$ for any $A, B \in S_n(\mathbb{R})$ and $\alpha, \beta \in \mathbb{R}$. Hence $S_n(\mathbb{R})$ is a subspace of $M_n(\mathbb{R})$.

For $1 \leq i \leq j \leq n$, define the matrix A_{ij} whose (ij) -th and (ji) -th entries are 1, and elsewhere 0. We claim that $\mathcal{B} = \{A_{ij} : 1 \leq i \leq j \leq n\}$ is a basis for $S_n(\mathbb{R})$.

First, for any $X = [x_{ij}] \in S_n(\mathbb{R})$, it is clear that $X = \sum_{i \leq j} x_{ij} A_{ij}$. Again, equating each entry of the matrix equation $\sum_{i \leq j} x_{ij} A_{ij} = \mathbf{O}$, we find that $x_{ij} = 0$ for all $1 \leq i \leq j \leq n$. Thus $\mathcal{B}$ is a basis for $S_n(\mathbb{R})$.

Notice that $\mathcal{B}$ has $\frac{n(n+1)}{2}$ elements. Hence $\dim(S_n(\mathbb{R})) = \frac{n(n+1)}{2}$. □
